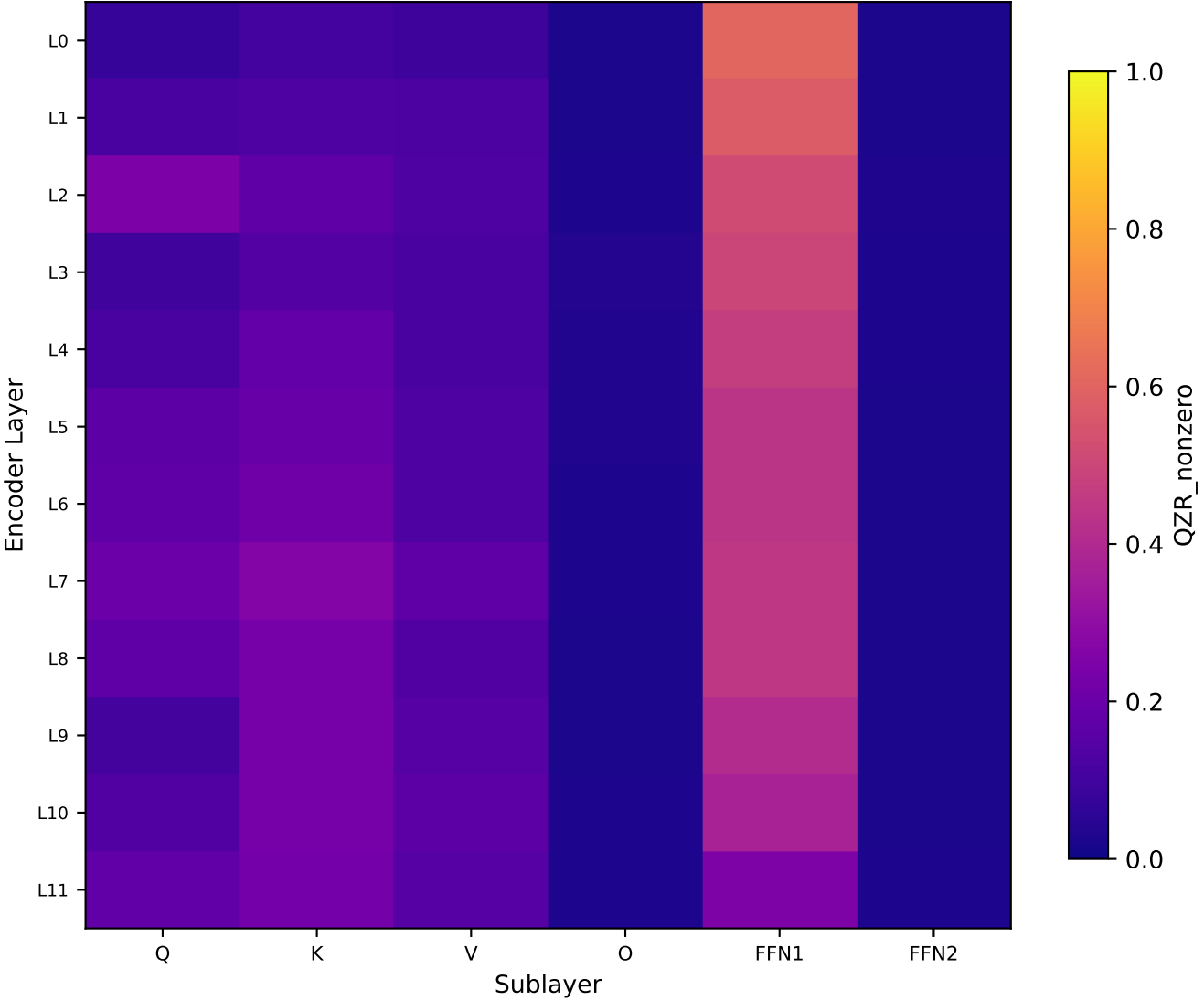
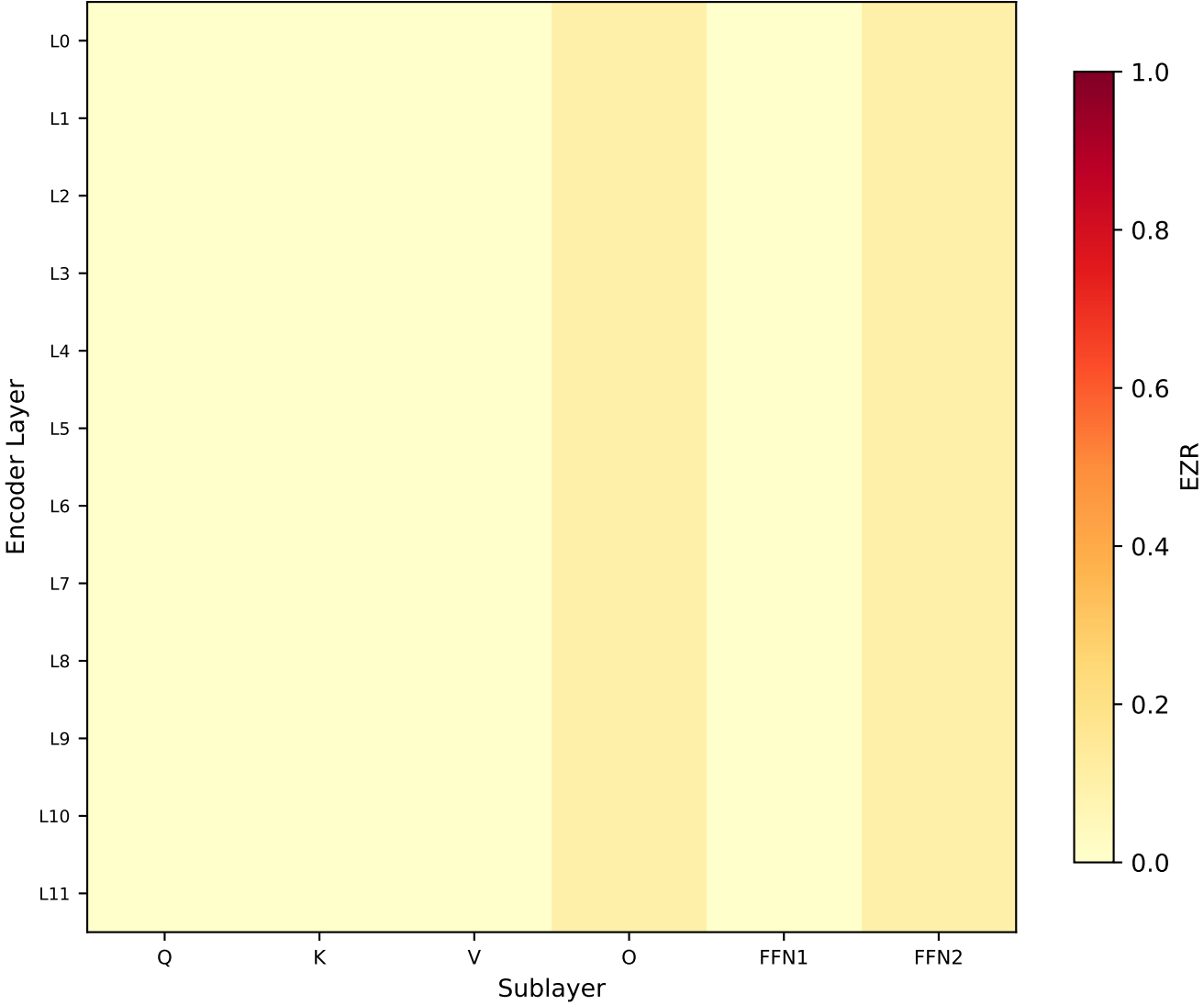


Figure A — QKVO+FFN Backward Root Cause Diagnosis (BERT-base, 5 steps × batch 2)
DAC=7-bit, zero_thresh=1/126=0.00794, 6 sublayers (QKVO+FFN1+FFN2)

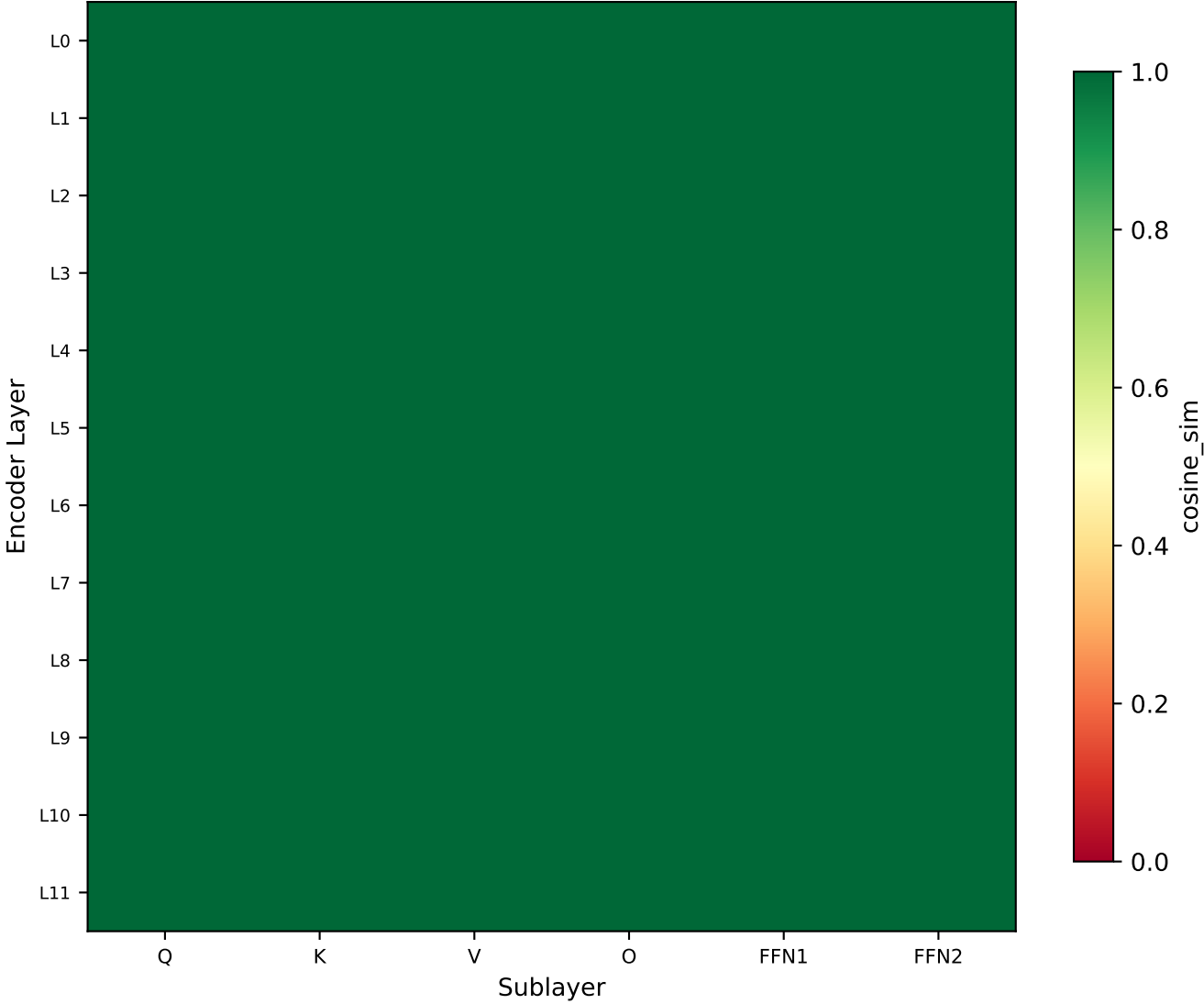
(a) QZR_nonzero (mask-excluded)
∈[0,1]: high = bulk-tiny/outlier-dom



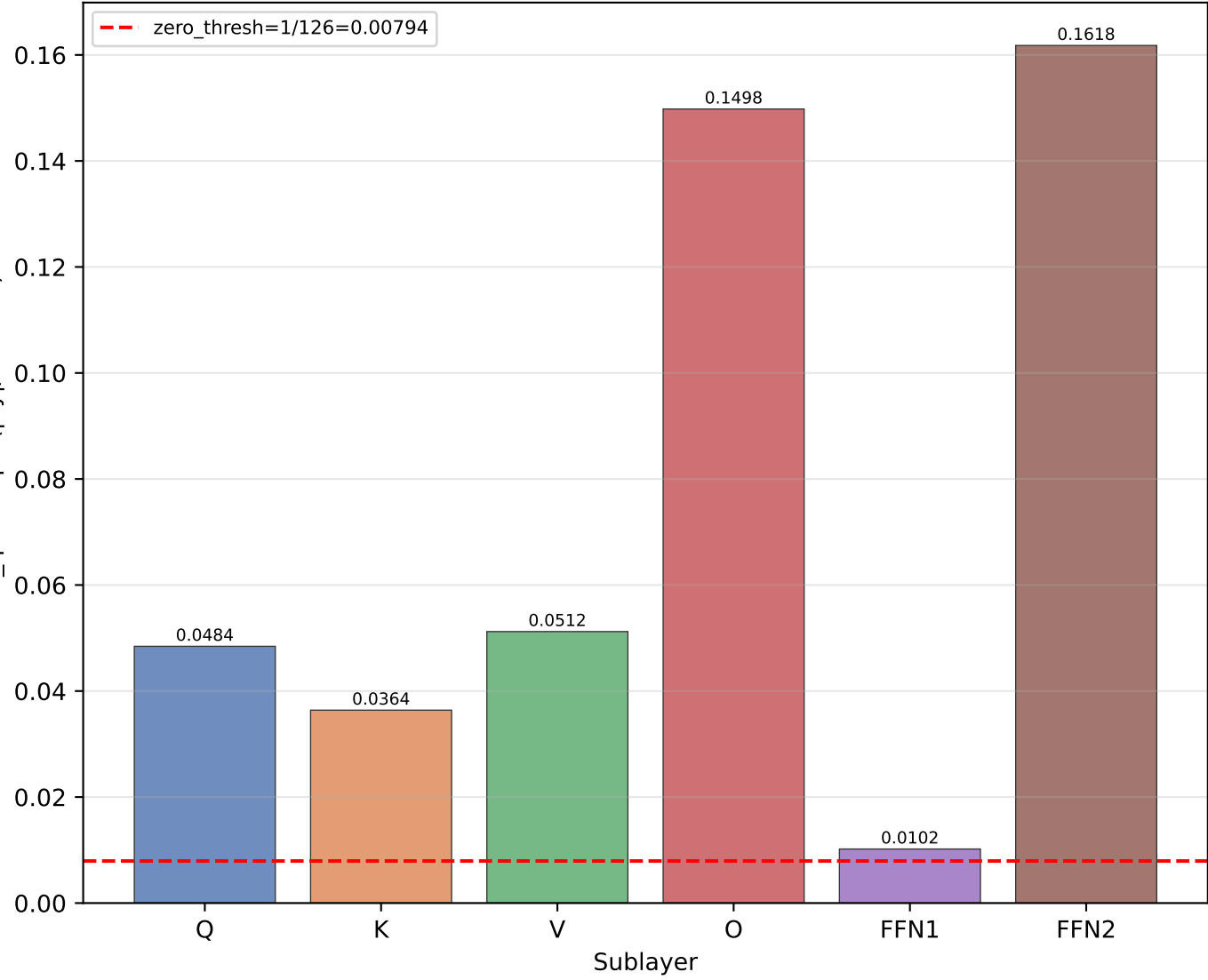
(b) EZR — Exact Zero Ratio (mask-excluded)
high = structural sparsity



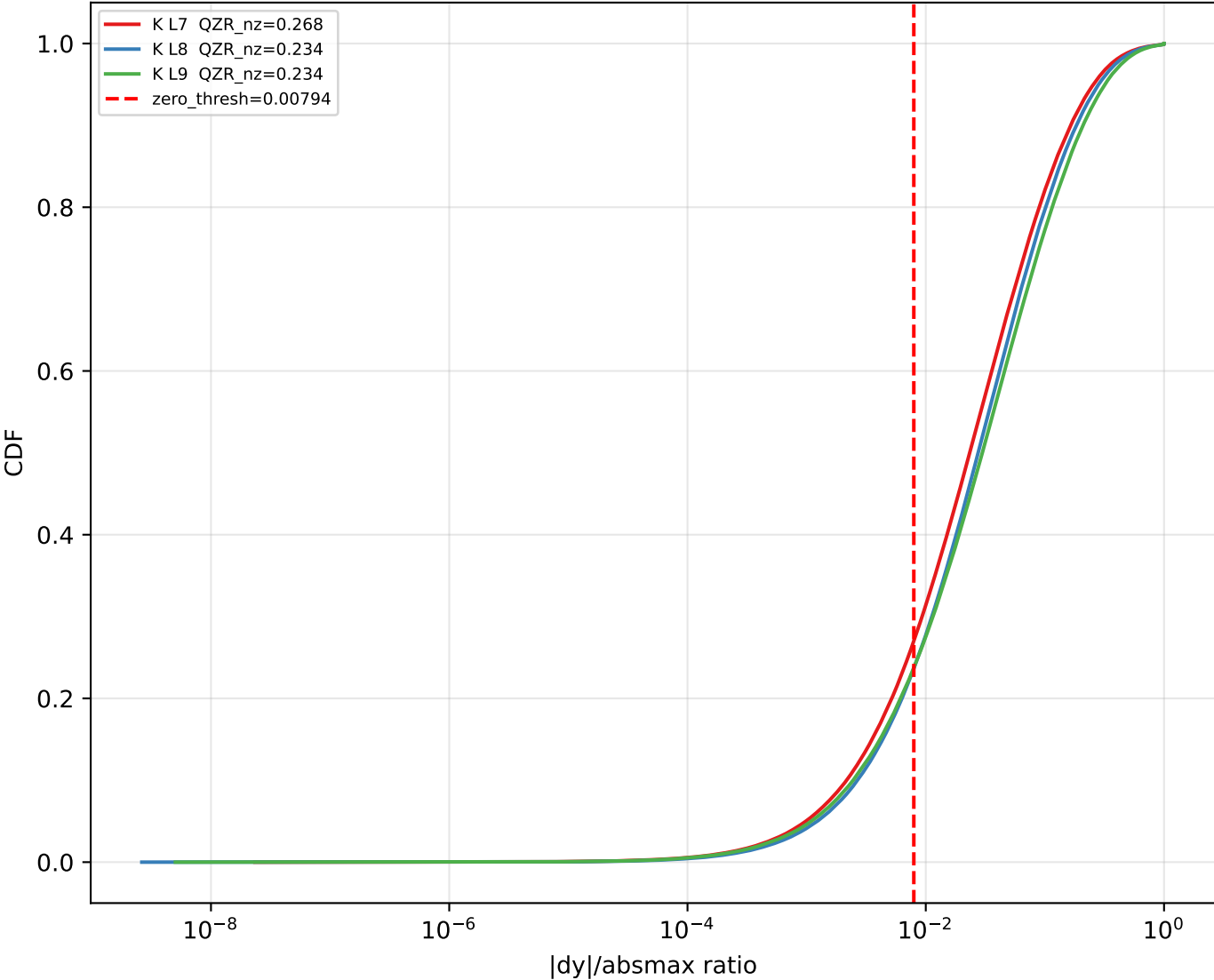
(c) Cosine Similarity FP32 vs DAC-quantized
(mask-excluded, higher = better)



(d) Median ratio per sublayer (mask-excluded)
ratio < zero_thresh → quantized to 0



(e) CDF of |dy|/absmax — worst 3 K layers
left of threshold → quantized to zero



(f) Diagnosis Panel

Auto-Diagnosis Summary	
=====	
DAC bits:	7
inp_res:	1/126 = 0.007937
zero_thresh:	0.007937
K/V EZR:	0.0000
K/V QZR_all:	0.1661
K/V QZR_nz:	0.1661
K/V cosine:	0.9999
K/V r_q50:	0.043806
FFN EZR:	0.0506
FFN QZR_nz:	0.2396
FFN cosine:	0.9997
□□ (K/V):	
□□□: EZR=0.00%,	
QZR_nz=16.61%	
ratio_q50=0.043806	